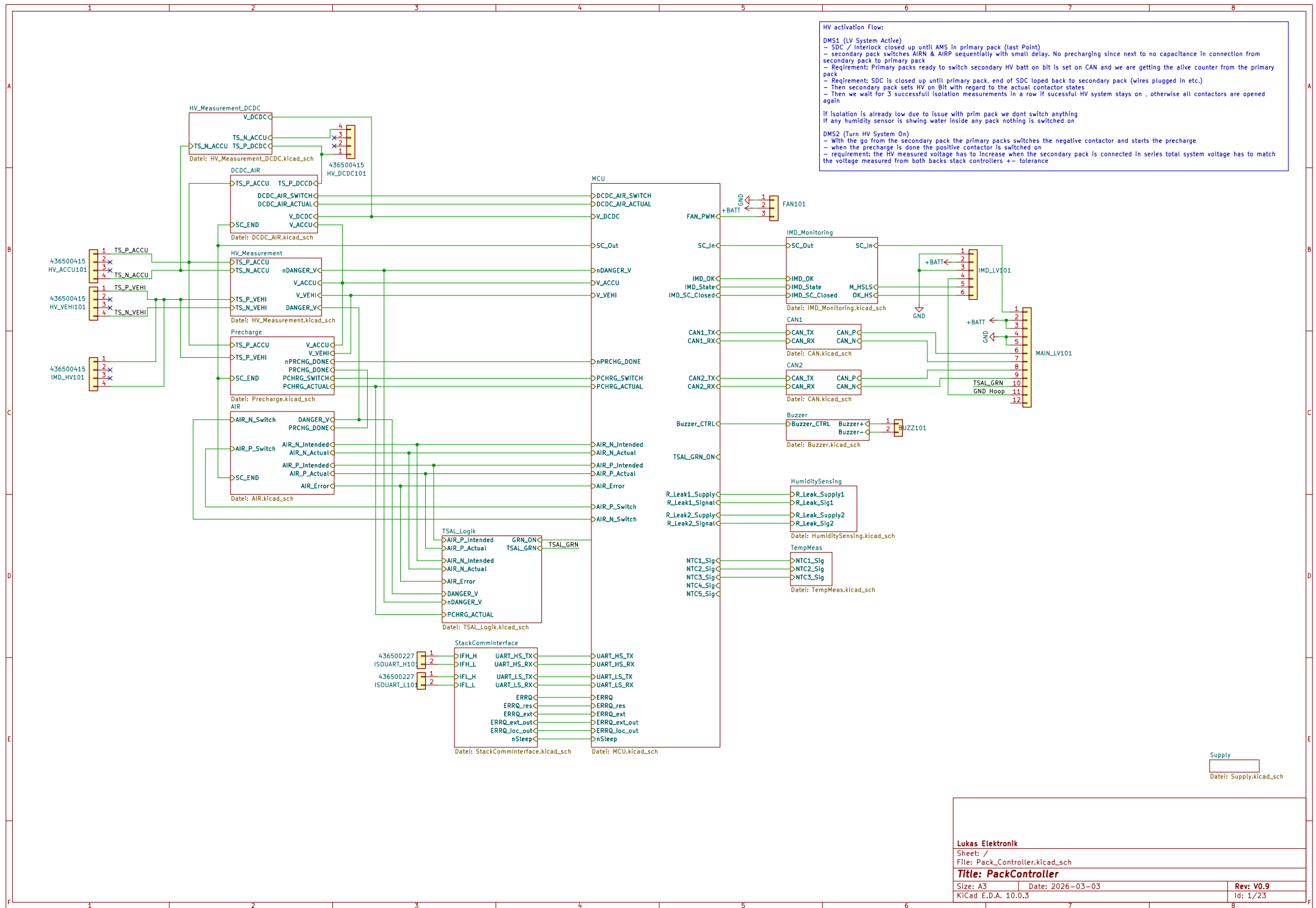




**HV activation Flow:**

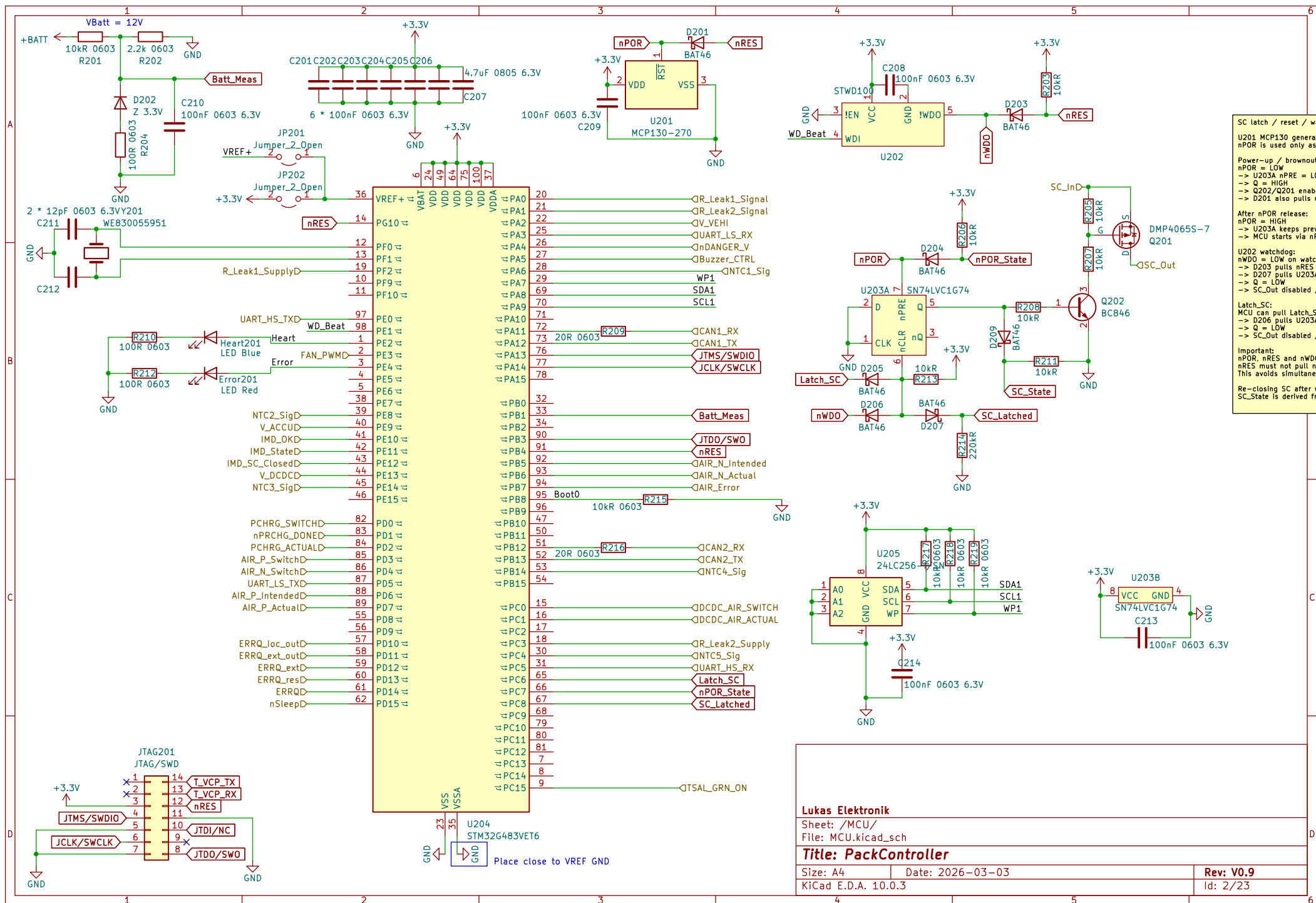
**DMS1 (LV System Active)**

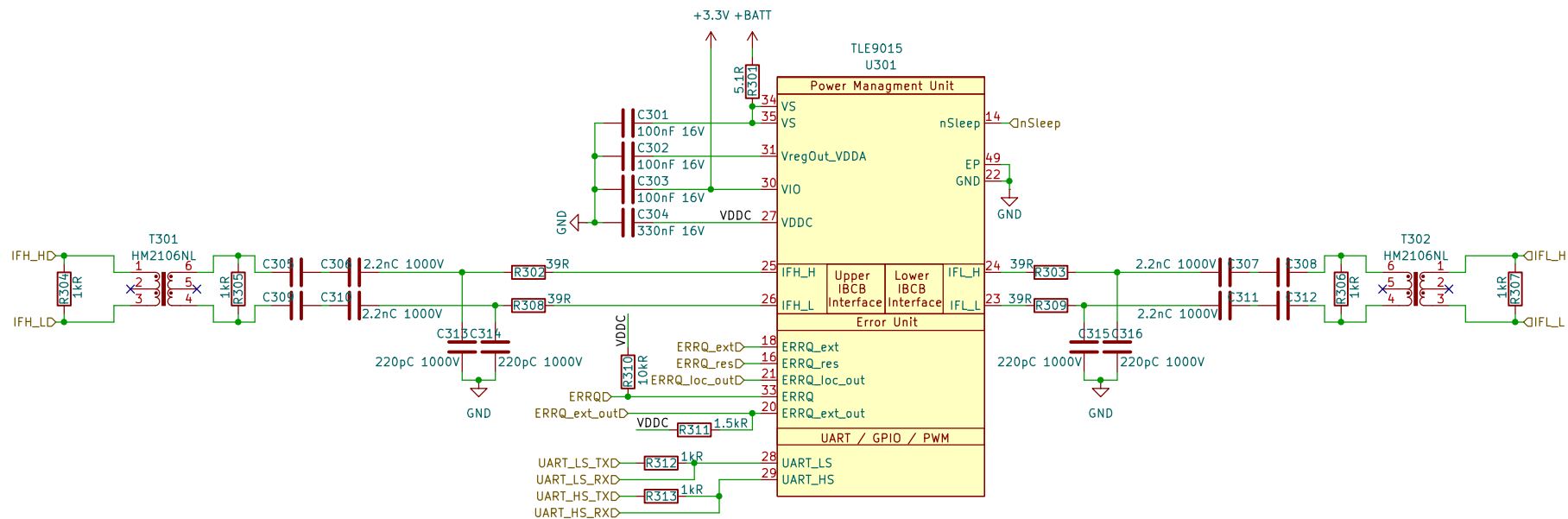
- SDC / Interlock closed up until AMS in primary pack (last Point)
- secondary pack switches AIRN & AIRP sequentially with small delay. No precharging since next to no capacitance in connection from secondary pack to primary pack
- Requirement: Primary packs ready to switch secondary HV batt on bit is set on CAN and we are getting the alive counter from the primary pack
- Requirement: SDC is closed up until primary pack, end of SDC loped back to secondary pack (wires plugged In etc.)
- Then secondary pack sets HV on BH with regard to the actual contactor states
- Then we wait for 3 successful isolation measurements in a row if successful HV system stays on, otherwise all contactors are opened again

If isolation is already low due to issue with prim pack we dont switch anything  
If any humidity sensor is shwing water inside any pack nothing is switched on

**DMS2 (Turn HV System On)**

- With the go from the secondary pack the primary packs switches the negative contactor and starts the precharge
- when the precharge is done the positive contactor is switched on
- requirement: the HV measured voltage has to increase when the secondary pack is connected in series total system voltage has to match the voltage measured from both backs stack controllers +/- tolerance





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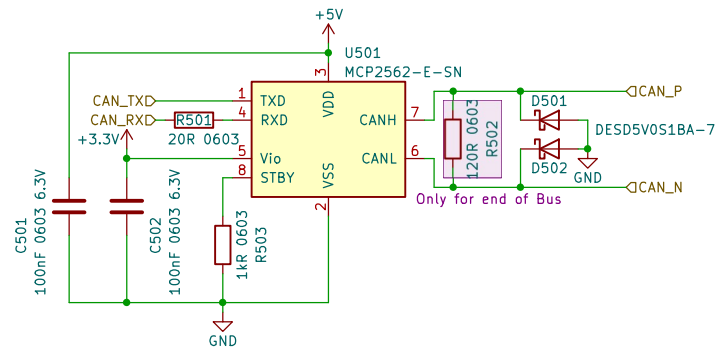
Sheet: /StackCommInterface/  
File: StackCommInterface.kicad\_sch

**Title: PackController**

Size: A4 Date: 2026-03-03  
KiCad E.D.A. 10.0.3

Rev: V0.9  
Id: 3/23





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Sheet: /CAN1/  
File: CAN.kicad\_sch

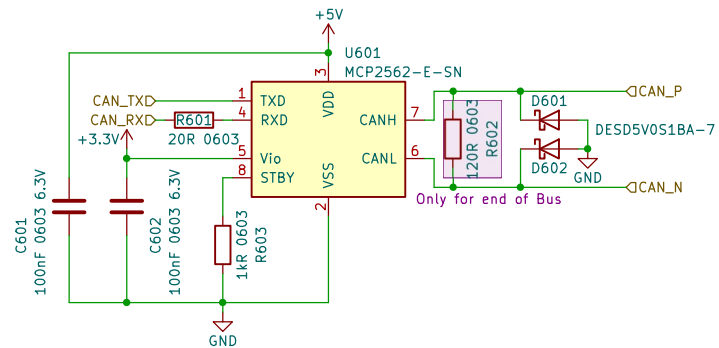
**Title: PackController**

Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

**Rev: V0.9**

Id: 5/23



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Sheet: /CAN2/  
File: CAN.kicad\_sch

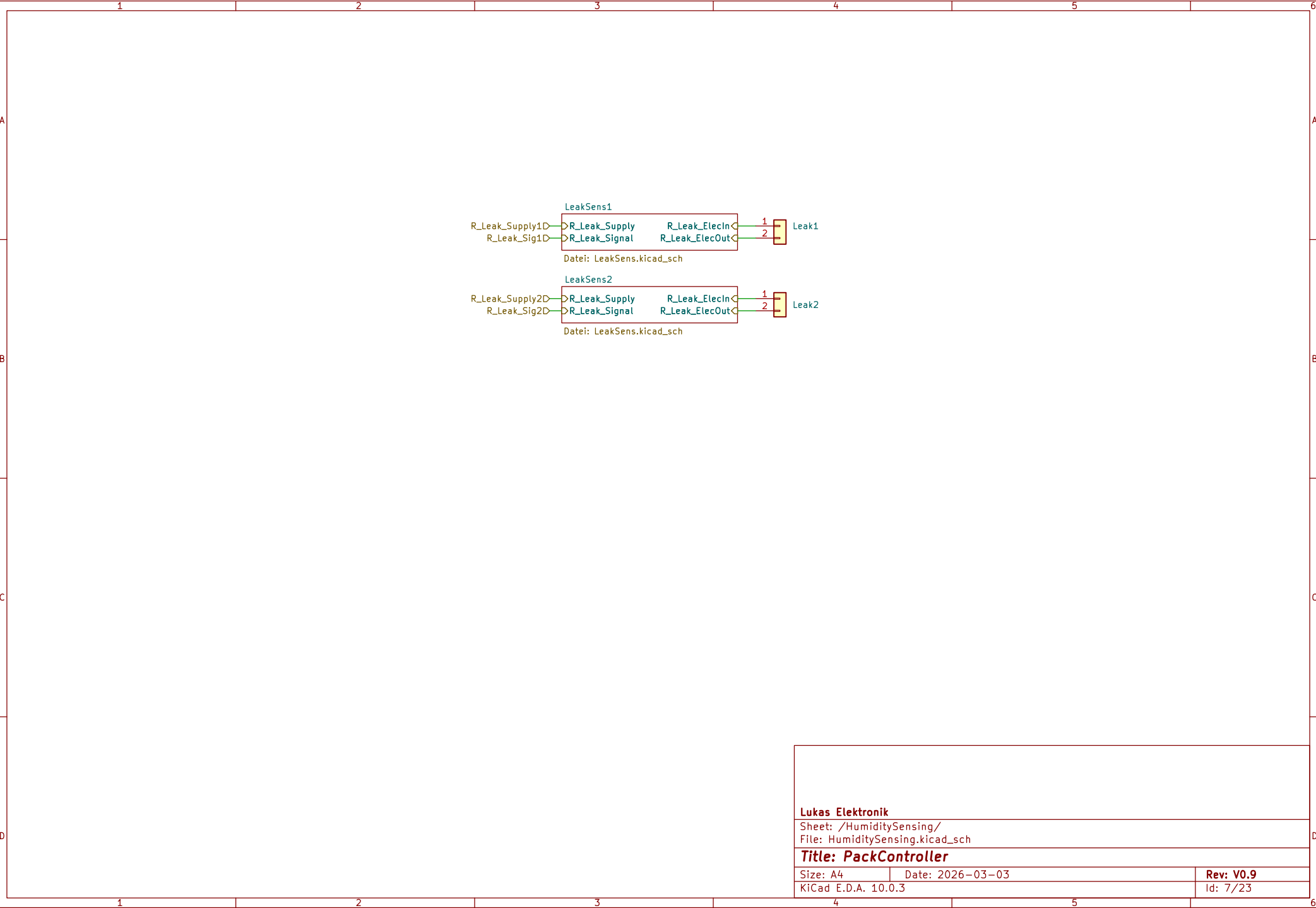
**Title: PackController**

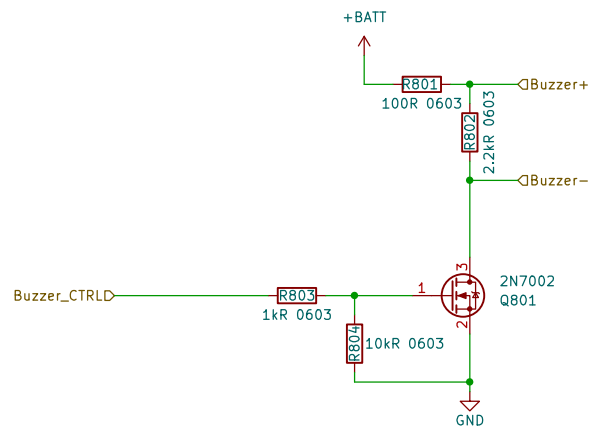
Size: A4 Date: 2026-03-03

KiCad E.D.A. 10.0.3

**Rev: V0.9**

Id: 6/23





Passive piezo buzzer driver.  
Target part: CPT-2305-90PM, externally driven, 12 Vp-p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:  
 $f_{\text{PWM}} \approx 4 \text{ kHz}$   
duty  $\approx 50 \%$

Q15 is a low-side NMOS switch.  
R37 discharges the piezo during MOSFET off-time:  
 $\tau = R37 * C_{\text{PIEZO}} \approx 2.2k * 24nF \approx 53\mu s$ .

R36 limits peak current and EMI.  
No additional series coupling capacitor required.

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Sheet: /Buzzer/

File: Buzzer.kicad\_sch

**Title: PackController**

Size: A4

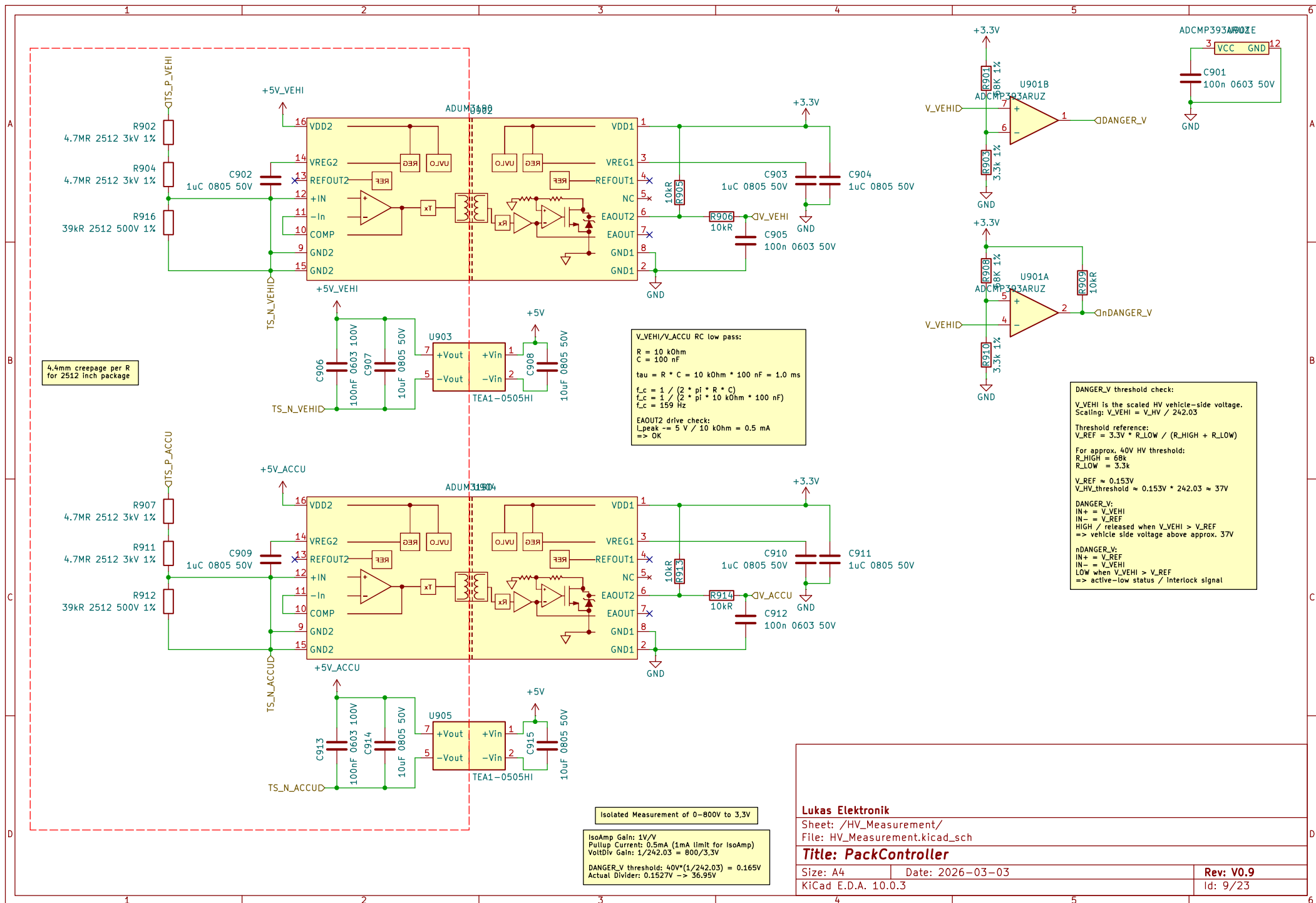
Date: 2026-03-03

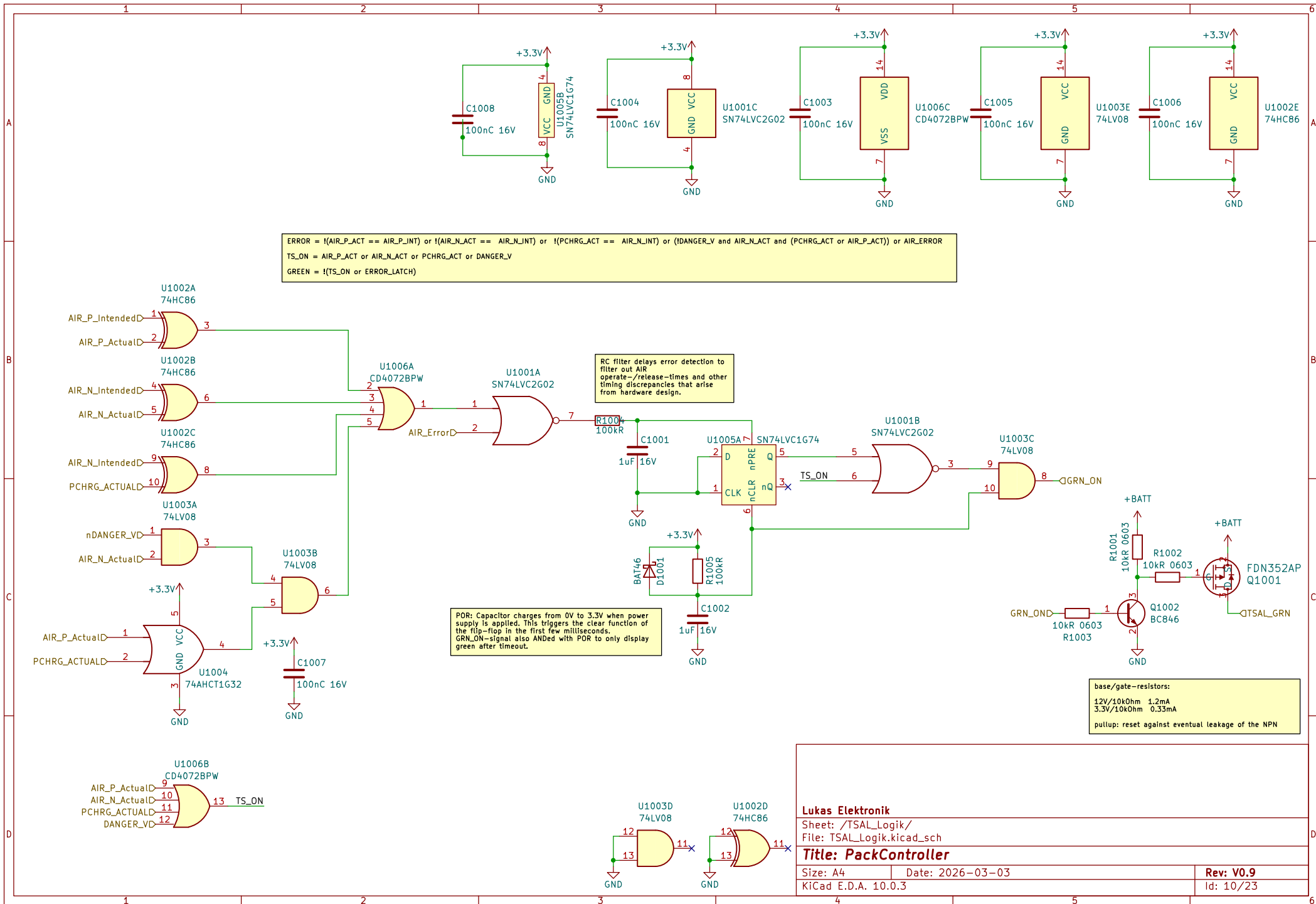
Rev: V0.9

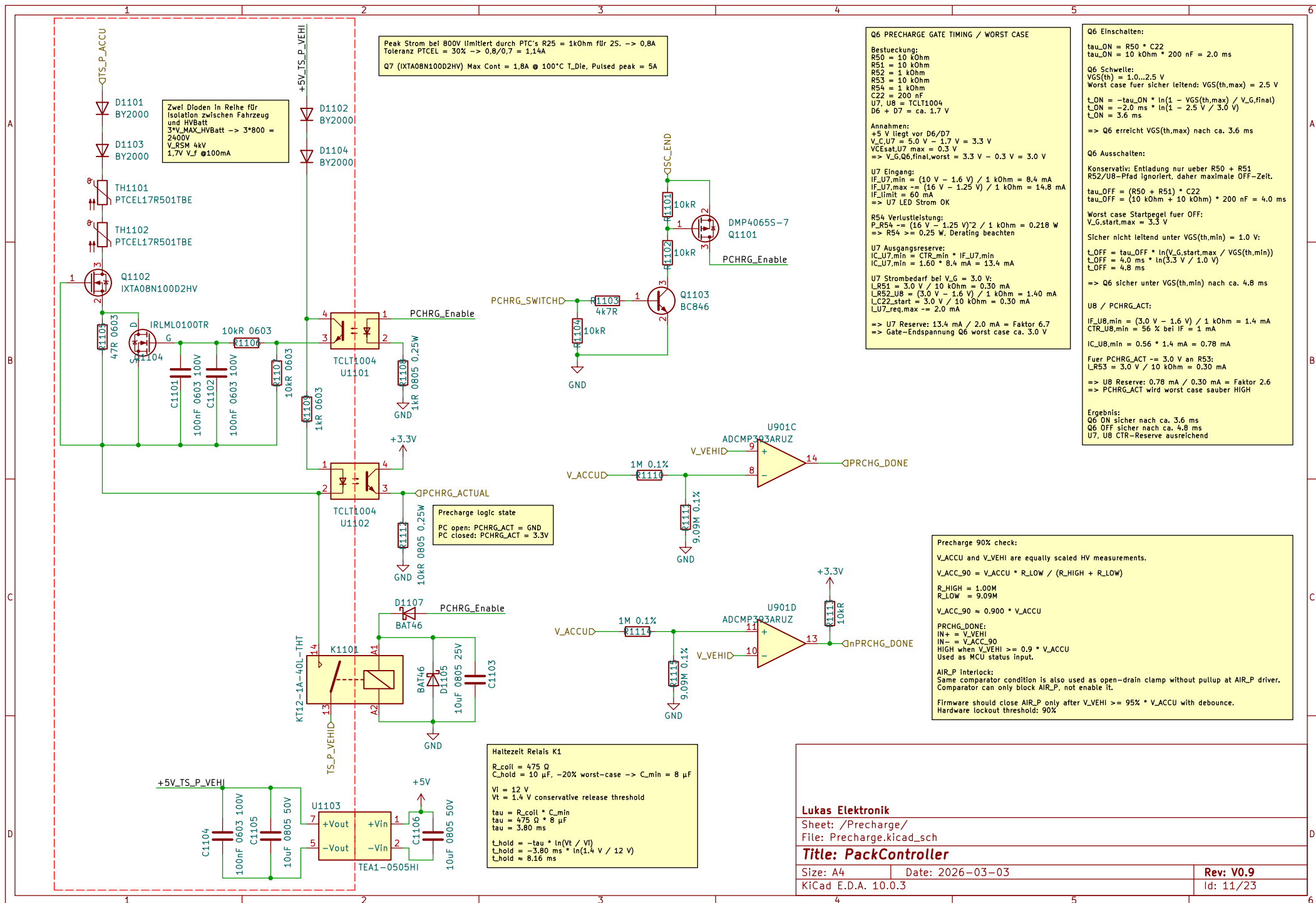
KiCad E.D.A. 10.0.3

Id: 8/23

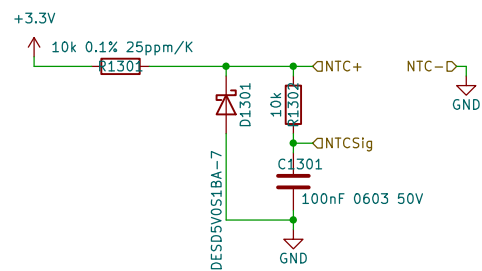












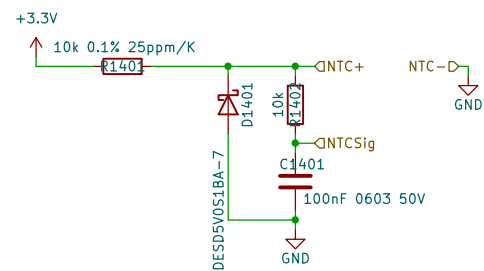
External NTC input:  
10k NTC to GND, 10k 0.1% pullup to 3V3.  
ADC is ratiometric to 3V3.

$R_{NTC} = R_{PULLUP} * x / (1 - x)$   
 $x = ADC_{raw} / ADC_{max}$

R58/C49 form ADC input filter:  
 $\tau = 10k * 100nF = 1ms$ ,  $f_c \approx 159Hz$ .

Open NTC -> ADC near full scale.  
Short to GND -> ADC near 0.

|                        |                  |           |
|------------------------|------------------|-----------|
| Lukas Elektronik       |                  |           |
| Sheet: /TempMeas/NTC1/ |                  |           |
| File: NTC1.kicad_sch   |                  |           |
| Title: PackController  |                  |           |
| Size: A4               | Date: 2026-03-03 | Rev: V0.9 |
| KiCad E.D.A. 10.0.3    |                  | Id: 13/23 |



External NTC input:  
10k NTC to GND, 10k 0.1% pullup to 3V3.  
ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$
$$x = ADC_{raw} / ADC_{max}$$

R58/C49 form ADC input filter:  
 $\tau = 10k * 100nF = 1ms$ ,  $f_c \approx 159Hz$ .

Open NTC -> ADC near full scale.  
Short to GND -> ADC near 0.

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Sheet: /TempMeas/NTC2/

File: NTC1.kicad\_sch

Title: PackController

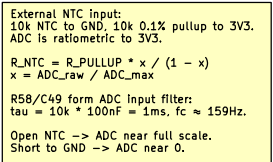
Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 14/23



Id: 15/23

Leakage sensor:  
Flex electrodes form R\_LEAK. R\_EOL = 10M is mounted on the flex sensor,  
parallel to the electrodes, for open-wire detection.

$R\_SENSOR = R\_LEAK || R\_EOL$

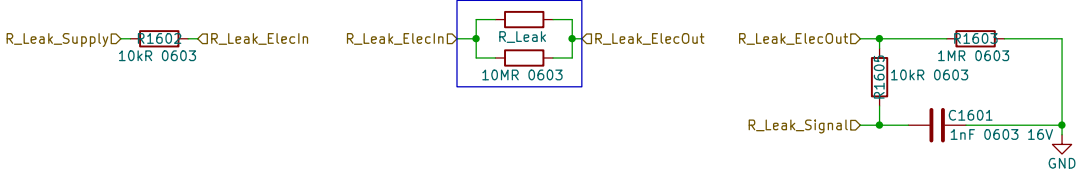
Measurement:  
Pulse R\_Leak\_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.  
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R\_Leak\_Supply = VDD:  
 $x = ADC\_raw / ADC\_max$   
 $R\_SENSOR = R\_REF * (1/x - 1) - R\_SER\_SUPPLY$

Nominal:  
R\_SER\_SUPPLY = 10k  
R\_REF = 1M  
R\_ADC\_SER = 10k  
C\_ADC = 1nF  
R\_EOL = 10M on flex

Expected:  
disconnected/open: ADC ≈ 0  
dry connected: R\_SENSOR ≈ 10M  
wet/leakage: R\_SENSOR decreases

Initial thresholds, validate with real coolant:  
warning: R\_SENSOR < 3M  
leak: R\_SENSOR < 1M  
severe leak: R\_SENSOR < 300k





Leakage sensor:  
Flex electrodes form R\_LEAK. R\_EOL = 10M is mounted on the flex sensor,  
parallel to the electrodes, for open-wire detection.

$R\_SENSOR = R\_LEAK || R\_EOL$

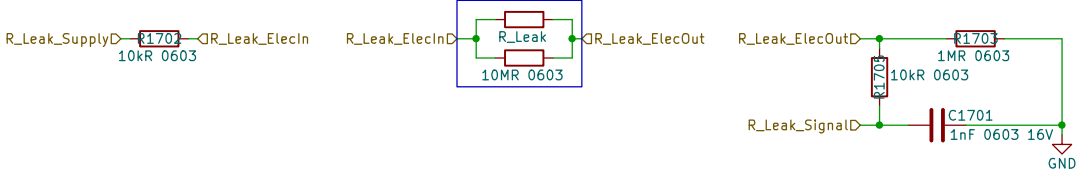
Measurement:  
Pulse R\_Leak\_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/HI-Z.  
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R\_Leak\_Supply = VDD:  
 $x = ADC\_raw / ADC\_max$   
 $R\_SENSOR = R\_REF * (1/x - 1) - R\_SER\_SUPPLY$

Nominal:  
R\_SER\_SUPPLY = 10k  
R\_REF = 1M  
R\_ADC\_SER = 10k  
C\_ADC = 1nF  
R\_EOL = 10M on flex

Expected:  
disconnected/open: ADC ≈ 0  
dry connected: R\_SENSOR ≈ 10M  
wet/leakage: R\_SENSOR decreases

Initial thresholds, validate with real coolant:  
warning: R\_SENSOR < 3M  
leak: R\_SENSOR < 1M  
severe leak: R\_SENSOR < 300k

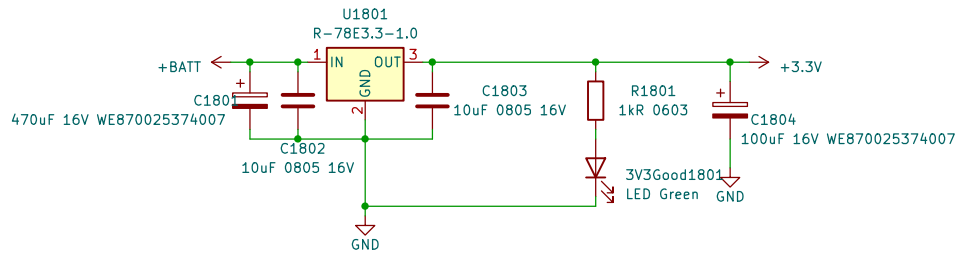


**+3V3 Supply worst-case estimate:**

Main loads:  
STM32G483 active incl. peripherals: -50 mA  
ADuM3190 VDD1, 2x: -4 mA  
24LC256 EEPROM write: -3 mA  
MCP2562 VIO, 2x worst-case: -1 mA  
ADCMP393 comparators + logic: <1 mA  
Pullups, dividers, GPIO drivers, LEDs: -20...30 mA

Estimated worst-case load: -80...90 mA  
Selected supply: RECOM 3.3 V / 1 A DC/DC

Design margin:  
1 A / 90 mA ≈ 11x

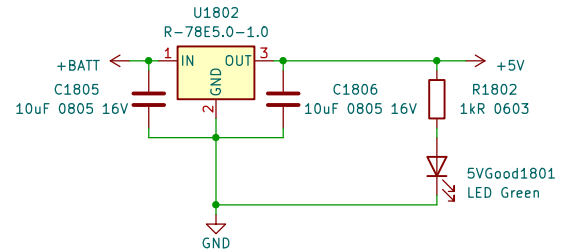


**+5V Supply worst-case estimate:**

Main loads:  
TEA1-0505HI Isolated DC/DC no-load, 3x: -84 mA  
TEA1 loads incl. ADuM3190 Isolated sides: -25...30 mA  
MCP2562 CAN transceivers, 2x dominant: -140 mA  
Misc. reserve / startup / tolerance: -50 mA

Estimated worst-case load: -250...300 mA  
Selected supply: RECOM 5 V / 1 A DC/DC

Design margin:  
1 A / 300 mA ≈ 3.3x



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Sheet: /Supply/  
File: Supply.kicad\_sch

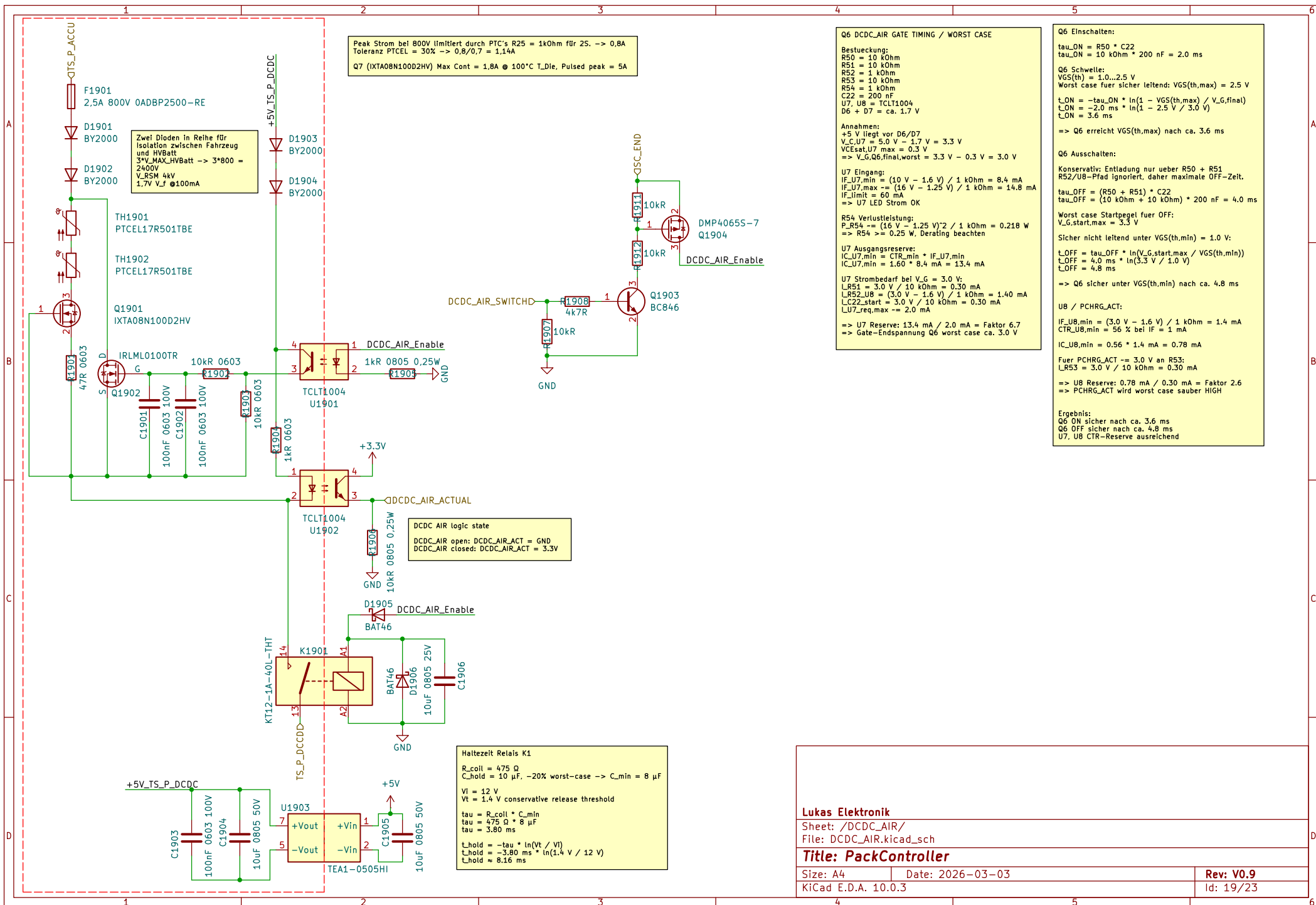
**Title: PackController**

Size: A4 Date: 2026-03-03

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**Rev: V0.9**

Id: 18/23



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Sheet: /DCDC\_AIR/

File: DCDC\_AIR.kicad\_sch

Title: PackController

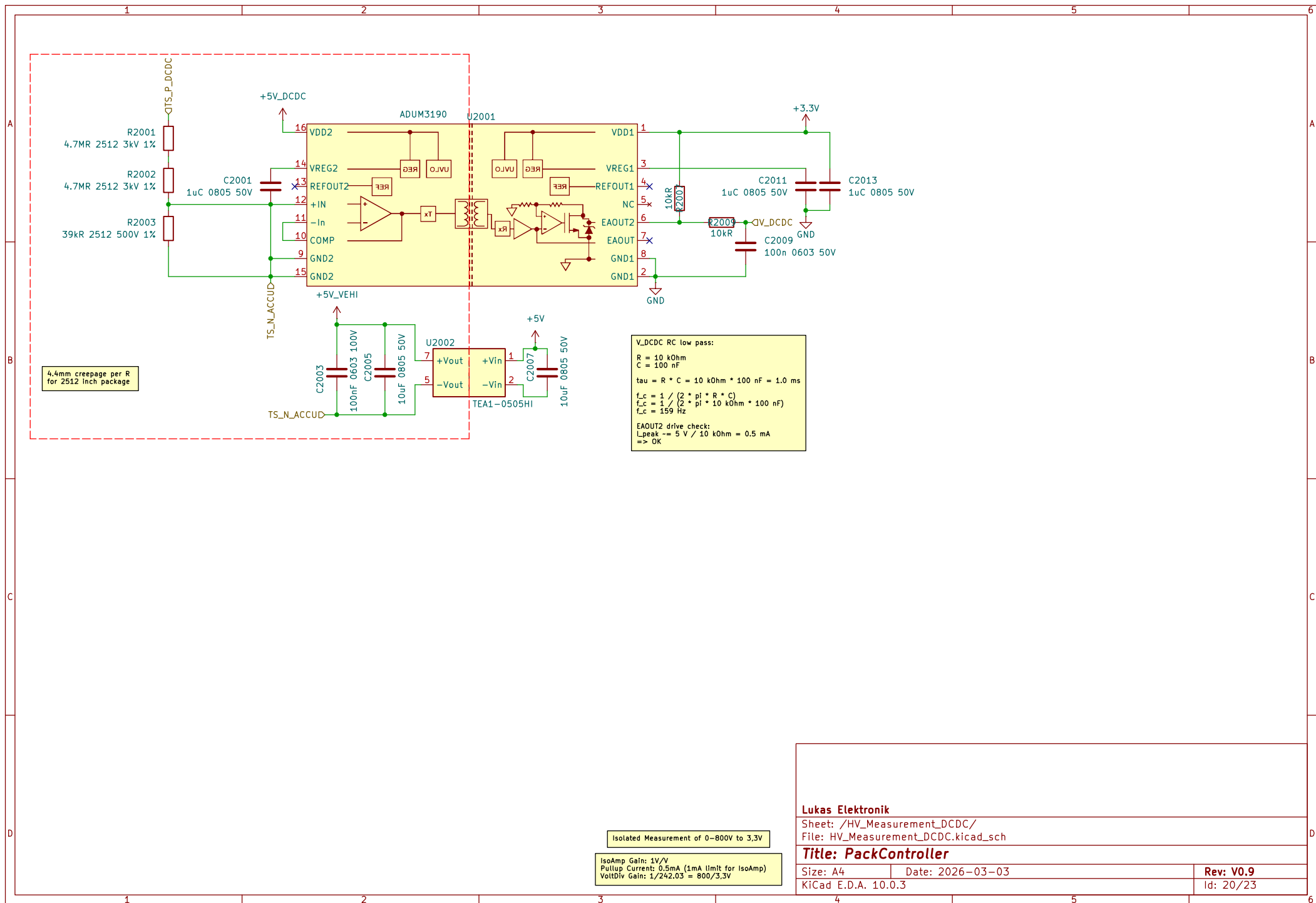
Size: A4

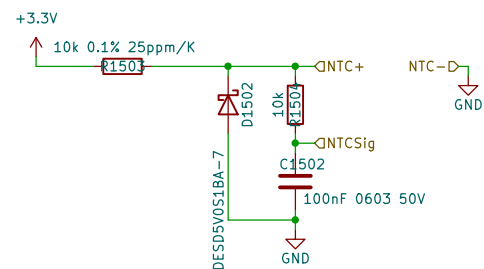
Date: 2026-03-03

KiCad E.D.A. 10.0.3

Rev: V0.9

Id: 19/23





External NTC input:  
 10k NTC to GND, 10k 0.1% pullup to 3V3.  
 ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$

$$x = ADC_{raw} / ADC_{max}$$

R5B/C49 form ADC input filter:  
 $\tau = 10k * 100nF = 1ms$ ,  $f_c \approx 159Hz$ .

Open NTC -> ADC near full scale.  
 Short to GND -> ADC near 0.

**Lukas Elektronik**

Sheet: /TempMeas/NTC4/

File: NTC1.kicad\_sch

**Title: PackController**

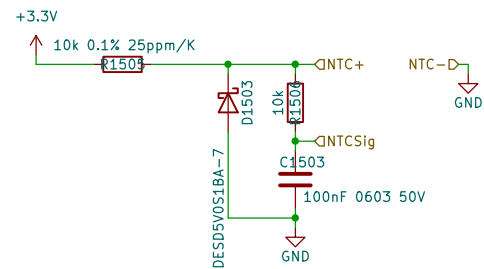
Size: A4

Date: 2026-03-03

**Rev: V0.9**

KiCad E.D.A. 10.0.3

Id: 21/23



External NTC input:  
 10k NTC to GND, 10k 0.1% pullup to 3V3.  
 ADC is ratiometric to 3V3.

$$R_{NTC} = R_{PULLUP} * x / (1 - x)$$

$$x = ADC_{raw} / ADC_{max}$$

R5B/C49 form ADC input filter:  
 $\tau = 10k * 100nF = 1ms$ ,  $f_c \approx 159Hz$ .

Open NTC -> ADC near full scale.  
 Short to GND -> ADC near 0.

**Lukas Elektronik**

Sheet: /TempMeas/NTC5/  
 File: NTC1.kicad\_sch

**Title: PackController**

Size: A4

Date: 2026-03-03

Rev: V0.9

KiCad E.D.A. 10.0.3

Id: 22/23

